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Proposal of Novel Image Compression Algorithmusing Kmean Clustering for VDI in Cloud Computing

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***Abstract*— In virtual desktop infrastructure (VDI) environments, the remote display protocol has a big responsibility: to transmit video data from a data-center-hosted desktop to the endpoint. In order to guarantee the best image quality and reduce the video delay time before encoding. Based on K-means clustering, we describe a novel compression algorithm of 2D images. Through simulations with a real-world dataset collected by the MIT Computer Science & Artificial Intelligence Lab, we show the experimental result. The compressed size is reduced significantly while ensuring the highest quality impossible.**

Keywords—vdi; mobile cloud computing; image processing; kmeans clustering;

I. INTRODUCTION

Cloud computing has become a hot topic in today's world, largely due to the flexibility and cost savings it can deliver. In the same manner as with visualization, cloud computing implementation started with server infrastructure and has moved to the desktop; the cloud is now ripe for desktop infrastructure. By moving desktops to the cloud rather than an internally deployed and managed VDI solution in a data center, businesses can realize all of the promised benefits of virtual desktops. There are many case studies in fact need to transmit data from the virtual desktop in the data cloud center to the screen endpoints, bring great meaning to the daily life. For example, in health care case, it is inconvenient for the doctor to check the patient status frequently. So, the idea is the screen contains information about heart rate or medical sign of specialized equipment will be transmitted to doctors who is observing the patient using their mobile devices which is connecting server by using remote desktop protocol [1]. This solution helps save valuable time and satisfies the demand for personal control as well as lower cost of long term medical care because it avoids the waste cost of spending human resources for manual processing [2]. Disease is always the leading cause of death that is why the new developments in health care are seriously concerned. In the other case, computer games have become very popular, gamers spent a lot of money on computer games, hardware, and accessories. Traditionally, computer games are delivered either in boxes or via Internet

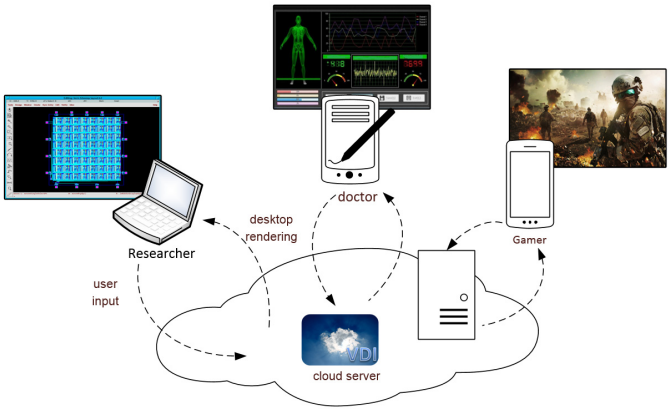

Fig. 1. General the model of desktop rendering between client and cloud.

download. Gamers have to install the computer games on physical machines to play them. The installation process becomes extremely tedious because the games are too complicated and the computer hardware and system software are very fragmented. Take Blizzard's Star-craft II as example, it may take more than an hour to install it on an i5 PC, and another hour to apply the online patches. Furthermore, gamers may find their computers are not powerful enough to enable all the visual effects yet achieve high frame rates. So, gamers have to repeatedly upgrade their computers so as to play the latest computer games. Hence, cloud gaming is a better way to deliver high-quality gaming experience and opens new business opportunity [3]. In a cloud gaming system, computer games run on powerful cloud servers and streaming video data to thin clients by using remote protocols [4]. The thin clients are light-weight and can be ported to resource-constrained platforms, such as mobile devices and TV set-top boxes. With cloud gaming, the key benefit is that gamers can play the latest computer gamers anywhere and anytime, while the game developers can optimize their games for a specific PC configuration. One more case study for using remote desktop protocols, in the laboratory, to perform complex tasks which required high CPU, resources or using specialized software. Instead of spending all that money for the license fee, the

researcher just need buy the software one time. All processing done at the cloud server and then delivery the screen content data back to the researcher.

From these case studies, compression technology is one of the most important contributions to improve the real-time video streaming quality in cloud VDI deployment scenario. By using K-means clustering algorithm [5], we propose a novel compression algorithm for 2D image to reduce the size of video data whenever transmitted over the network while ensuring the highest quality possible.

The rest of this paper is organized as follows. Section II presents the novel image compression algorithm based on K-means. The experiments and results are demonstrated in Section III. Finally, conclusion is given in Section IV.

II. PROPOSED IMAGE COMPRESSION ALGORITHM USING K-MEANS CLUSTERING

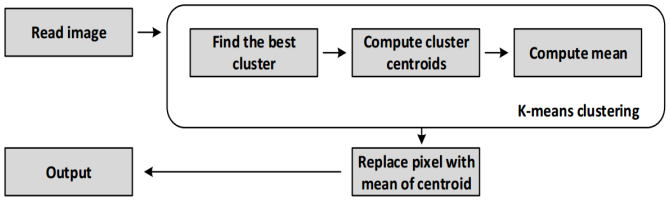

Fig. 2. Image compression flowchart

2D images typically use either 8-bit or 24-bit color. When using 8-bit color, there is a definition of up to 256 colors forming a palette for this image, each color denoted by an 8-bit value. A 24-bit color scheme, as the term suggests, uses 24 bits per pixel and provides a much better set of colors. For our purposes we will consider 24 bit RGB (R'G'B') to be uncompressed color. The number 24 refers to the total number of bpp (bits per pixel) used to describe color. A 24 bit number has a range of 0 (no color or black) to 16,777,215 for a total of 16,777,216 different colors. Since Red, Green, and Blue are each described by 8 bits (24/3), which gives a range of 0 (no Red, no Green, or no Blue) to 255 for a total of 256 possible variations of each primary color. From this we can start to get an idea that the real-image contains thousands of colors and reduce the number of colors as a suitable number of colors. It can be 8, 16 or 32 colors, depend on the user demand and the network condition which is present in the next section.

By making this reduction, it is possible to represent (compress) the photo in an efficient way. For example, suppose we want to reduce the number of colors to 16 of 256x256 image. Specifically, we need to store the RGB values of the 16 selected colors, and for each pixel in the images we now need to only store the index of the color at that location (where only 4 bits are necessary to represent 16 possibilities). Concretely, we will treat every pixel in the original image as a data example and use the k-means algorithm to find the 16 colors that best group (cluster) the pixels in the 3 dimensional RGB space. Once we have computed the cluster centroids on the image and then assign each pixel position to its closet centroid. After that, we can view the effects of the compression by reconstructing the image based on the centroid assignments. Specifically, replace each pixel location with the mean of

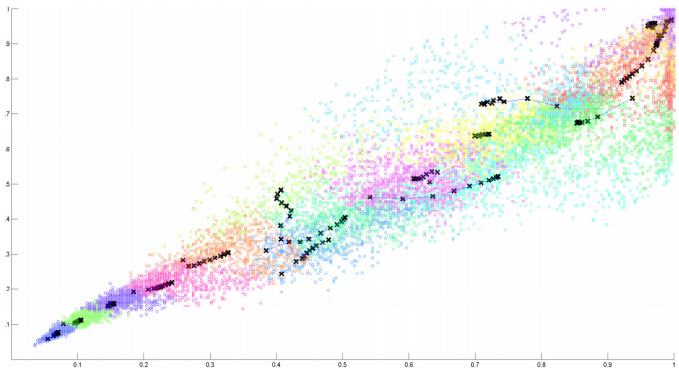

Fig. 3. The centroids stopped moving in the 10th interation, a convergence termination criterion is met. 16 colors (cluster) that best group be created.

centroid assigned. Eventually, notice that after these steps we have significantly reduced the number of bits that are required to describe the image. The results can be summarized as follows: The original image required 24 bits for each one of the 256x256 pixel locations, so total size of 256 x 256 x 24 = 1,572,864 bits. The new representation requires some overhead storage in form of a dictionary of 16 colors, each of which require 24 bits, but the image itself then only requires 4 bits per pixel location. The final number of bits used is therefore 16 x 24 + 256 x 256 x 4 = 262,528 bits, which corresponds to compressing the original image by about a factor of 6.

Input: Real image
Output: Reconstructed image

Steps:
1. Load the color image A(m x n);
2. Read every pixel in the original image as sample data;
3. Apply K-means algorithm to find the colors (cluster) that best group pixels in the 3 dimensional RGB space;
4. Compute the cluster centroids again;
5. Assign each pixel position to its closet centroid;
6. Compute the mean of all the centroids;
7. Replace each pixel location with the mean of centroid assigned;

Algorithm. 1. Image compression algorithm using Kmeans clustering.

III. EXPERIMENTAL REULT

The proposed method was tested on some images in the dataset collected by the MIT Computer Science & Artificial Intelligence Lab [6]. The natural images in the set include Butterfly Zebra (300×212, 27.4 KB), Street Boston (590×430, 069.2 KB), Conference Room (600 × 450, 61.4 KB), Airport (745 × 258, 46.2 KB), Outdoor Oxford (1526 × 2048, 1.21 MB), Chip Designer (1680 × 1050, 503 KB) and Angry Bird (1920×1080, 153 KB). All of the image compression programs were implement in the Matlab 2010 and executed on Intel Core i5-4670 CPU @ 3.40GHz machine.

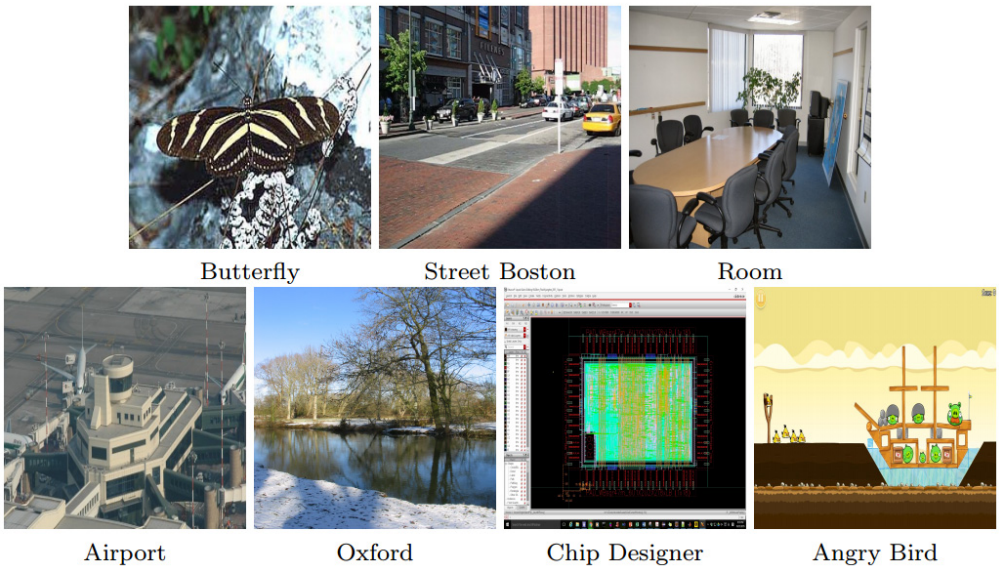

Fig. 4. Test images

The effectiveness of a compression method was quantified by the Mean Squared Error (MSE) measure:

$$MSE(X,\hat{X}) = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left\|x(h,w)-\hat{x}(h,w)\right\|_2^2 \quad (1)$$

Where $X,\hat{X}$ denote respectively the $H \times W$ original and compressed images in the RGB color space. MSE represent the average distortion with respect to the L_2^2 norm [7]. Note that the Peak Signal-to-Noise Ratio (PSNR) measure can be easily calculated from the MSE value:

$$PSNR = 20\log_{10}\left(\frac{255}{MSE}\right) \quad (2)$$

Let α denotes the number of color depth, image dimensions $A(m \times n)$, μ be defined as the number of colors (cluster) and γ as the number of bits are necessary to represent the index of the colors at the location in the image. We computed compression ratio (CR) by dividing the size of the original file in bits by the size of the compressed file:

$$CR = \frac{|G|}{\mu \times \alpha + m \times n \times \gamma} \quad (3)$$

where $|G| = \alpha \times m \times n$ denotes size of the original image. Figure 5 represents the original image Turtle and a compressed version of Turtle with $\mu = 32$, respectively. The results of compression with difference μ values are shown in Table 1 and Figure 6.

As can be seen that this approach has enable to compress an image to 70%, when the compressed image is visually 95% similar to the original image. Actually, the compressed size is reduced significantly while keeping the quality, still like on the original image.

a) Turtle.jpg – 514 KB

b) Turtle_32.jpg – 221 KB

Fig. 5. Comparison of compression image and the original with $\mu = 32$. The PSNR of compressed image is 26.5845 dB.

IV. CONCLUSION

In this paper, based on K-mean clustering algorithm, we proposed a novel compression method for the 2D image for VDI in cloud computing. The experimental was also presented as the video data size is significantly reduced when transmitted over the Internet while ensuring highest quality.

TABLE I. THE COMPRESSION SIZE, PSNR & COMPRESSION RATIO

	1. Buffterfly Zebra (300x212) - 27.4 KB				
	$\mu=8$	$\mu=16$	$\mu=32$	$\mu=64$	$\mu=128$
Size (KB)	21.7	20.6	19.8	19.8	19.7
PSNR (dB)	24.675	28.6469	30.7613	33.0554	34.6483
CS	1.2626	1.33	1.3838	1.3838	1.3908
	2. Street Boston (590x430) - 69.2 KB				
	$\mu=8$	$\mu=16$	$\mu=32$	$\mu=64$	$\mu=128$
Size (KB)	52.5	50.2	49.4	49.3	49
PSNR (dB)	24.815	28.0013	29.8641	31.9285	33.9174
CS	1.3180	1.3784	1.4000	1.4036	1.4122
	3. Conference Room (600x450) - 61.4 KB				
	$\mu=8$	$\mu=16$	$\mu=32$	$\mu=64$	$\mu=128$
Size (KB)	36.2	40.7	38.7	39.6	38
PSNR (dB)	25.217	28.7521	30.0919	30.0919	33.9716
CS	1.6961	1.5085	1.5865	1.5505	1.6157
	4. Airport (745x258) - 46.2 KB				
	$\mu=8$	$\mu=16$	$\mu=32$	$\mu=64$	$\mu=128$
Size (KB)	36.9	37.1	37.5	38.2	38.6
PSNR (dB)	30.274	31.6957	33.0064	34.7103	35.3905
CS	1.2520	1.2452	1.2320	1.2094	1.1968
	5. Oxford (1536x2048) - 1.21 MB				
	$\mu=8$	$\mu=16$	$\mu=32$	$\mu=64$	$\mu=128$
Size (KB)	653	612	619	603	589
PSNR (dB)	23.220	25.3275	28.3941	31.1382	32.5477
CS	1.8974	2.0245	2.0016	2.0547	2.1036
	6. Chip Designer (1680x1050) - 503 KB				
	$\mu=8$	$\mu=16$	$\mu=32$	$\mu=64$	$\mu=128$
Size (KB)	374	338	339	355	358
PSNR (dB)	20.100	22.5293	20.002	24.5301	27.4084
CS	1.3449	1.4881	1.4837	1.4169	1.4050
	7. Angry Bird (1920x1080) - 153 KB				
	$\mu=8$	$\mu=16$	$\mu=32$	$\mu=64$	$\mu=128$
Size (KB)	143	132	129	126	125
PSNR (dB)	20.645	25.3002	29.3269	30.8296	35.4472
CS	1.0699	1.1590	1.1860	1.2142	1.224

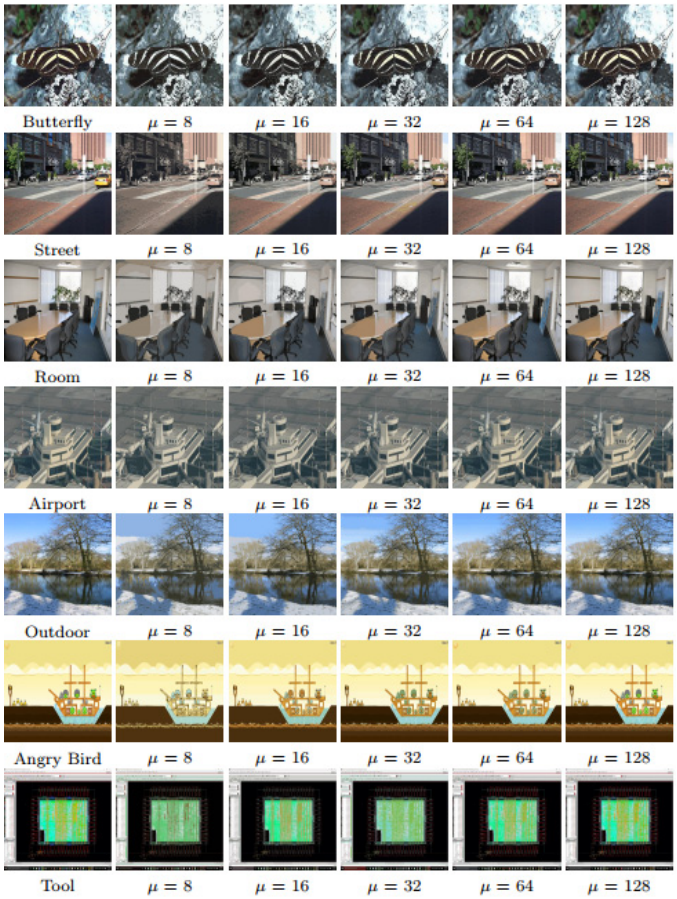

Fig. 6. The image results of compression with difference μ values.

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